CAMBRIDGE INSTITUTE OF TECHNOLOGY

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Department of Information Science & Engineering
Final Year B.E Project Work: 2026-27

Project Synopsis

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Guide name: Prof. Anusha.B Batch No: 39

Title: A Personalized Menstrual Health Framework for Phase-Specific Optimization of Diet, Physical Training, and Recovery Using Time-Series Modeling and Wearable Physiological Signals

ABSTRACT

The menstrual cycle's four hormonal phases — menstrual, follicular, ovulatory, and luteal — directly govern energy metabolism, neuromuscular performance, recovery kinetics, and nutritional demands in women athletes. Existing health platforms operate on phase-agnostic paradigms, yielding suboptimal performance outcomes and elevated injury risk.

This system addresses the gap through a phase-aware, multi-model intelligent framework. An LSTM network performs sequential cycle phase classification from longitudinal menstrual data, while an XGBoost module drives training intensity modulation and recovery scheduling based on physiological baselines and fatigue indicators. A nutritionist-validated, rule-augmented dietary engine dynamically adjusts macronutrient targets and micronutrient priorities per hormonal state. A fine-tuned LLM powers a conversational assistant for natural-language symptom logging, phase queries, and real-time guidance. RNN-based pattern recognition supports longitudinal health trend analysis, while wearable integration enables continuous physiological monitoring. Automated phase-transition notifications, alarm scheduling, and an analytics dashboard complete the pipeline.

Built on a React frontend with a FastAPI backend, PostgreSQL, and a hybrid rule-knowledge base, the system unifies hormonal biology, evidence-based sports nutrition, and adaptive intelligence into a cohesive decision-support platform purpose-built for female athletic performance.

SDGs matching our project:

SDG 3 – Good Health and Well-Being: This project promotes healthier training practices by helping women understand how their menstrual cycle affects their physical performance and nutritional needs. By providing data-driven insights, the system can reduce health risks such as fatigue, hormonal imbalance, and overtraining.

SDG 5 – Gender Equality: Sports science research and training plans are often designed based on male physiological patterns. This project addresses the gender gap by focusing specifically on women's biological cycles and providing personalized insights that support female athletes.

SDG 9 - Industry, Innovation, and Infrastructure: Builds innovative, accessible technology that bridges the gap between hormonal science and intelligent systems, an area largely overlooked in mainstream health tech. Its scalable, modular design makes it deployable across healthcare and sports industries, contributing directly to sustainable digital infrastructure.

SDG 10 – Reduced Inequalities: Access to sports science experts and personalized training advice is limited for many athletes. A digital platform that provides data-driven recommendations can help make this knowledge more accessible to women from diverse backgrounds.

Name of the Guide: Prof.Anusha.B Signature of Guide:
Date: 09/03/2026

Signature of Project Coordinator: